Customer Churn Forecasting

Predicting Customers Likely to Stop Using a Service

1. Project Objective

The objective is to develop a customer churn forecasting system that identifies customers who are likely to stop using a service. The solution combines data preprocessing, feature engineering, exploratory analysis, multiple classification models, class-imbalance handling, model evaluation, and feature-importance analysis to support proactive retention strategies.

2. Dataset Details

The project uses the IBM Telco Customer Churn dataset. The dataset contains 7,043 customer records and 21 fields describing demographics, tenure, subscribed services, contract type, billing/payment behavior, charges, and the Churn target.

Item	Value
Original records	7,043
Original features	21
Records after cleaning	7,043
Churn rate	26.54%
Training split	60%
Validation split	20%
Test split	20%

3. Data Preprocessing

The data was checked for duplicates, missing values, numeric consistency, and impossible values. TotalCharges required special handling because 11 records contained blank values. These were converted to numeric and imputed using MonthlyCharges × tenure. No duplicate customer rows were found. Negative tenure or charge values were excluded if present.

4. Feature Engineering

• TenureGroup: groups customers by service tenure.

• ServiceCount: counts subscribed service features.

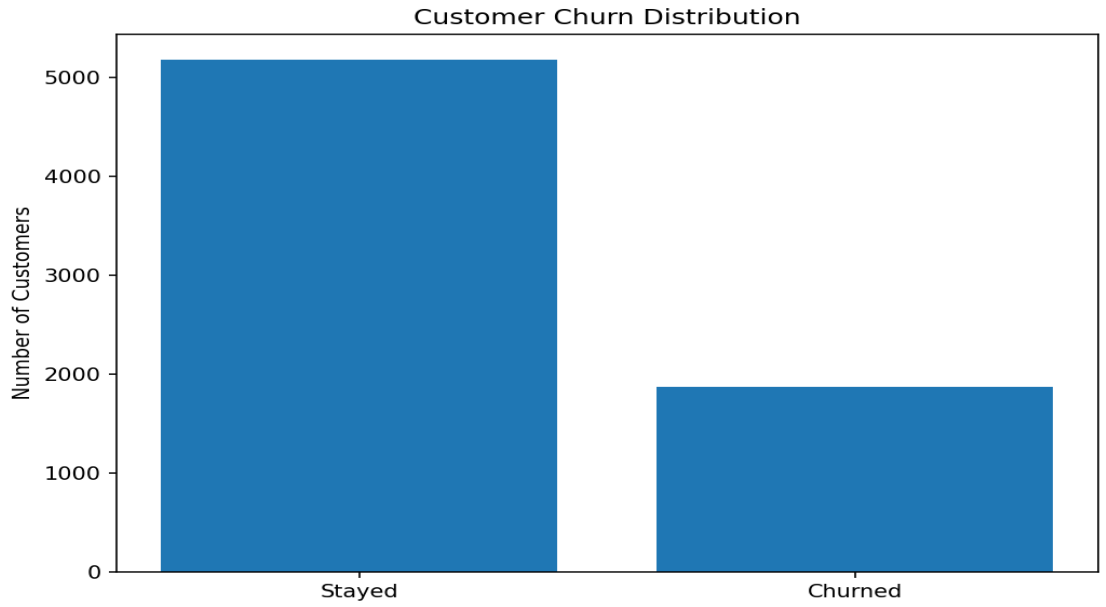
• AvgMonthlySpend: estimates average monthly spending from TotalCharges and tenure.

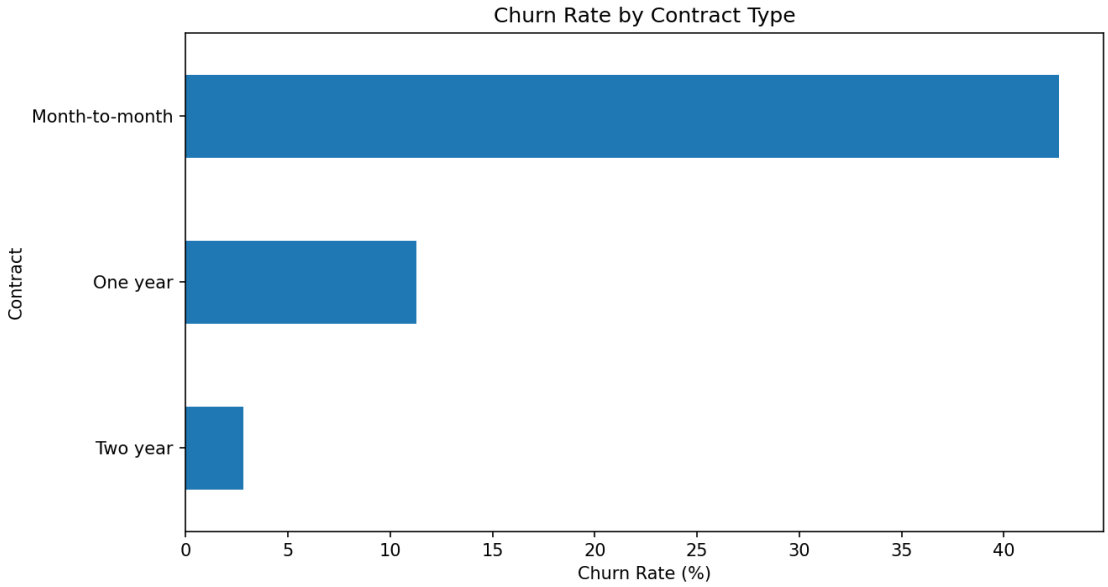
• IsMonthToMonth: identifies customers on month-to-month contracts.

• HasAutoPayment: identifies automatic bank-transfer or credit-card payment.

• HasInternetService: identifies customers with an internet service.

5. Exploratory Data Analysis





The overall churn rate is 26.54%. Contract type shows a large difference in observed churn: month-to-month customers have a churn rate of 42.71%, compared with 11.27% for one-year contracts and 2.83% for two-year contracts.

6. Class Imbalance

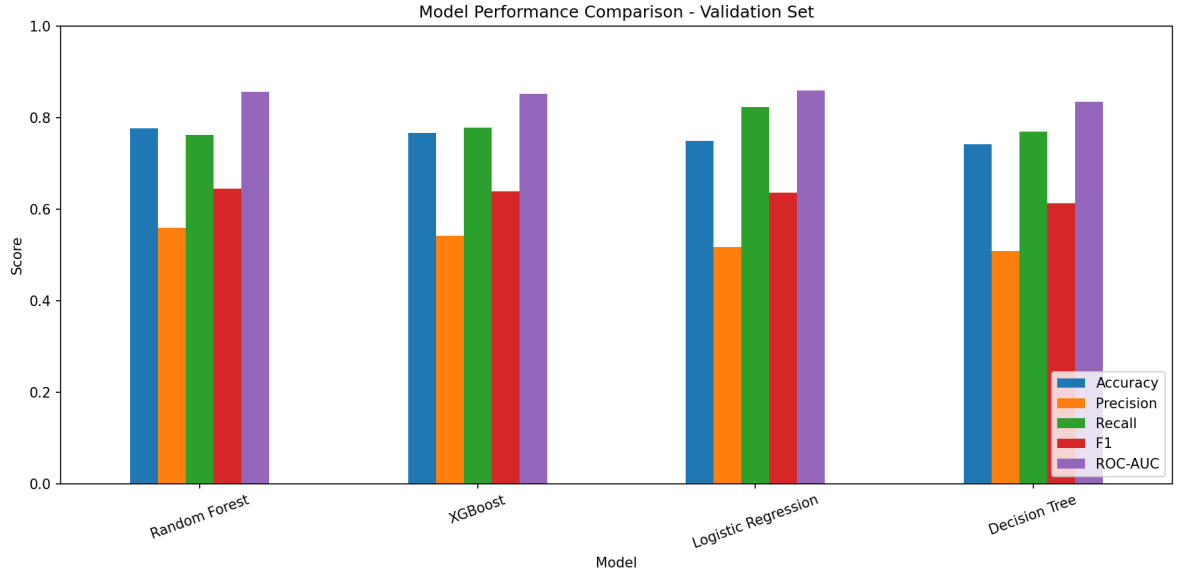
Churn is imbalanced because fewer customers churn than remain. To reduce the effect of this imbalance during model training, class_weight='balanced' was used for Logistic Regression, Decision Tree, and Random Forest. XGBoost used scale_pos_weight based on the training class ratio. Stratified train/validation/test splits preserved the churn proportion across datasets.

7. Model Development

Four classification approaches were evaluated where available: Logistic Regression, Decision Tree, Random Forest, and XGBoost. Numerical features were median-imputed and standardized, while categorical variables were imputed using the most frequent category and one-hot encoded. All preprocessing was placed inside scikit-learn pipelines to avoid data leakage.

8. Validation Results

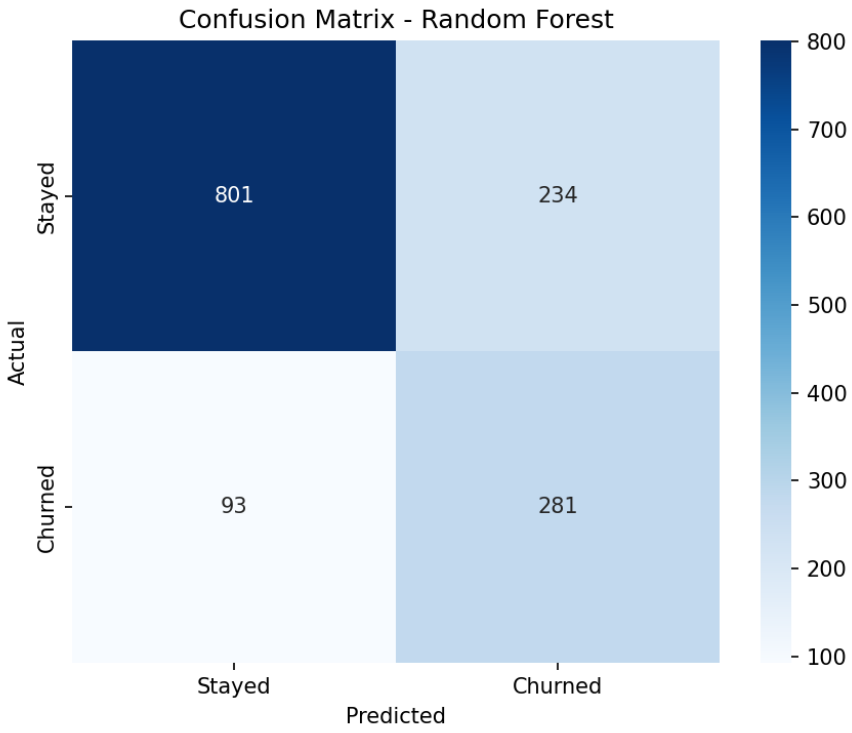
Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	0.777	0.559	0.762	0.645	0.857
XGBoost	0.767	0.542	0.778	0.639	0.852
Logistic Regression	0.749	0.518	0.824	0.636	0.859
Decision Tree	0.742	0.509	0.770	0.613	0.835



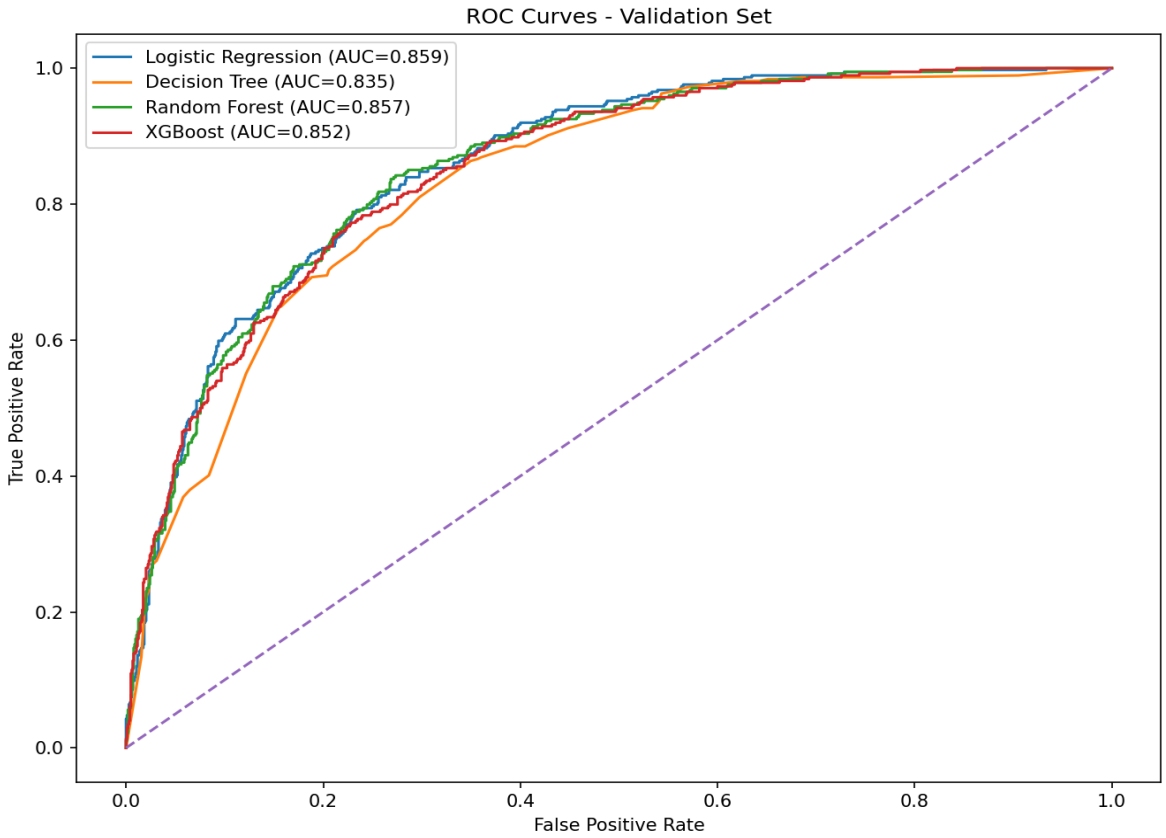
9. Final Test Performance

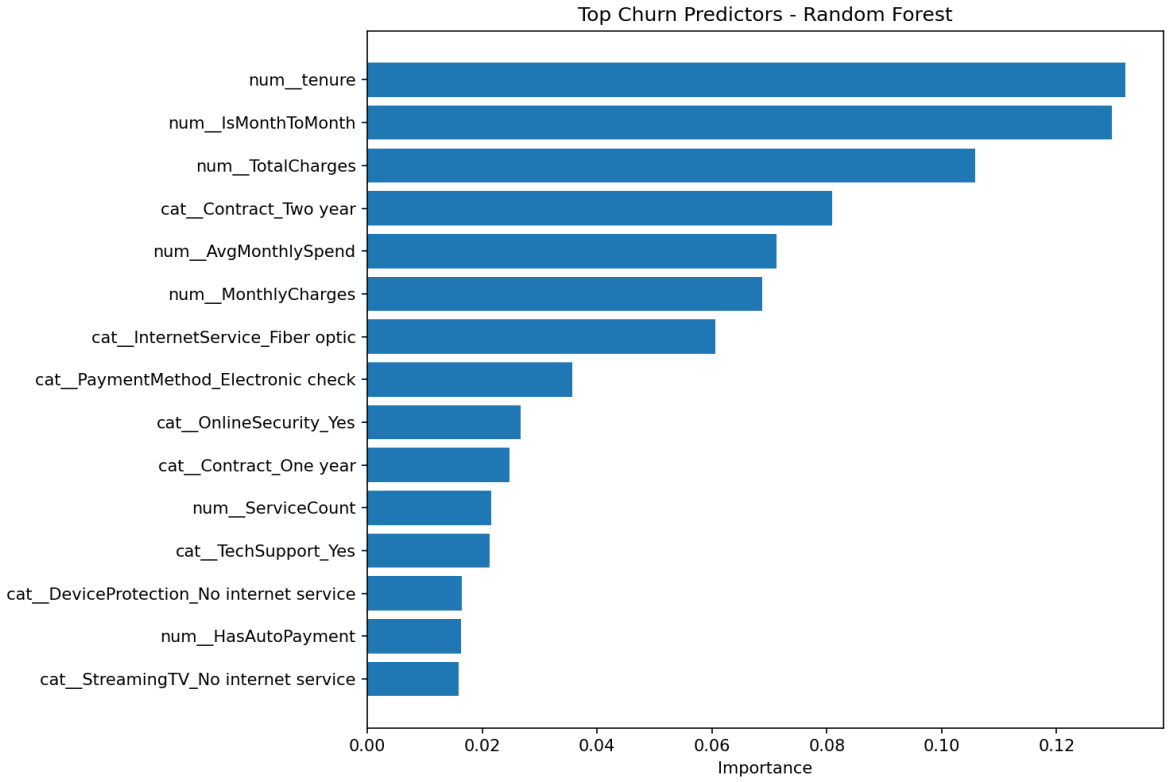
Metric	Test Score
Accuracy	0.7679
Precision	0.5456
Recall	0.7513
F1	0.6322
ROC-AUC	0.8312

The model selected using validation F1 was Random Forest. On the held-out test set it achieved accuracy 0.7679, precision 0.5456, recall 0.7513, F1 0.6322, and ROC-AUC 0.8312. Recall is especially relevant in churn prevention because missing a customer who is actually going to churn can reduce the opportunity for proactive retention.


Confusion Matrix - Random Forest
801
234
93
281
Stayed
Churned
Actual
Stayed
Churned
Predicted
800
700
600
500
400
300
200
100

10. ROC Curves and Feature Importance





11. Top Churn Predictors

- num__tenure: importance 0.1319
- num__IsMonthToMonth: importance 0.1296
- num__TotalCharges: importance 0.1059

• cat__Contract_Two year: importance 0.0809

• num__AvgMonthlySpend: importance 0.0712

• num__MonthlyCharges: importance 0.0687

• cat__InternetService_Fiber optic: importance 0.0606

• cat__PaymentMethod_Electronic check: importance 0.0357

• cat__OnlineSecurity_Yes: importance 0.0266

• cat__Contract_One year: importance 0.0247

12. Business Insights

• The dataset has an overall observed churn rate of 26.54%.

• Month-to-month customers show a churn rate of 42.71%, compared with 11.27% for one-year contracts and 2.83% for two-year contracts.

• Customers in the 0-6 month tenure group show the highest observed churn among the defined tenure groups, at approximately 52.94%.

• Customers using automatic payment methods have an observed churn rate of 15.98%, compared with 34.67% for customers without automatic payment.

• The Random Forest feature-importance analysis places tenure, month-to-month contract status, total charges, contract type, monthly spending, and internet-service characteristics among the leading predictors.

13. Actionable Retention Strategies

• Prioritize at-risk month-to-month customers for proactive retention campaigns.

• Create onboarding and early-life support programs for customers in the first months of tenure.

• Use churn scores to prioritize retention outreach instead of contacting every customer equally.

• Review pricing, service experience, and support for customers with higher monthly charges or specific service combinations.

• Promote longer-term contracts when appropriate, while evaluating customer needs and incentives.

• Monitor recall and precision together: increasing recall catches more potential churners, while precision controls unnecessary retention outreach.

14. Challenges and Limitations

The dataset is a historical customer snapshot, so the model identifies statistical patterns rather than proving that a factor causes churn. Feature importance should be interpreted as predictive contribution, not causal impact. Thresholds can also be adjusted depending on the business cost of false positives versus missed churners.

15. Conclusion

The completed churn forecasting system combines preprocessing, feature engineering, EDA, class-imbalance handling, multiple predictive models, validation, held-out testing, and feature-importance analysis. It provides a practical framework for identifying customers with higher churn risk and supporting targeted retention decisions.